Is Elementary AI Education Possible?

Abstract

As artificial intelligence (AI) technology becomes increasingly pervasive, it is critical that students recognize AI and how it can be used. There is little research exploring learning capabilities of elementary students and the pedagogical supports necessary to facilitate students' learning. PrimaryAI was created as a 3rd-5th grade AI curriculum that utilizes problem-based and immersive learning within an authentic life science context through four units that cover machine learning, computer vision, AI planning, and AI ethics. The curriculum was implemented by two upper elementary teachers during Spring 2022. Based on pre-test/post-test results, students were able to conceptualize AI concepts related to machine learning and computer vision. Results showed no significant differences based on gender. Teachers indicated the curriculum engaged students and provided teachers with sufficient scaffolding to teach the content in their classrooms. Recommendations for future implementations include greater alignment between the AI and life science concepts, alterations to the immersive problem-based learning environment, and enhanced connections to local animal populations.